Research questions from the current analysis

Multimodal drivers + TerraMind embeddings for daily soil moisture

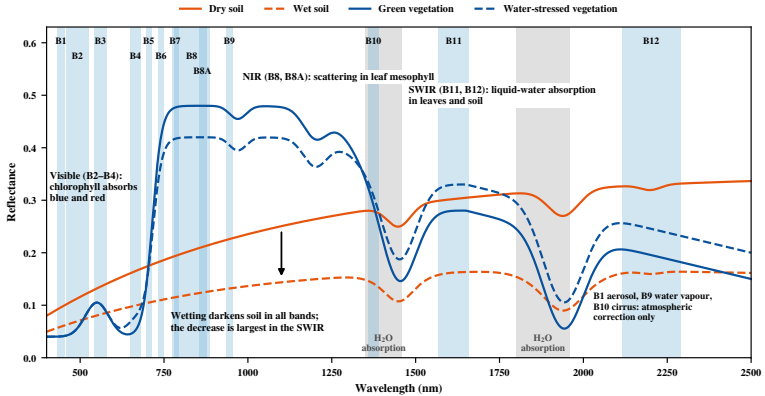
Prajwal Khanal

7 October 2026

Motivation Where does soil moisture information come from?

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\mathrm{SM}, \mathrm{roughness}, \mathrm{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\mathrm{SM} = f_\theta(\mathrm{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

Motivation What does a satellite see of soil moisture?



- Wetting darkens bare soil in every band, most in the SWIR (B11, B12)
- Vegetation hides the soil: NIR and SWIR respond to leaf water, not soil water
- The signal is indirect and mixed with land cover, texture and canopy
- No single band or index maps to soil moisture ⇒ learn the mapping from many modalities

Motivation Geospatial foundation models (GFMs)

What they are

- Large networks (usually Vision Transformers) pre-trained on huge *unlabelled* Earth-observation archives
- Self-supervised: learn by reconstructing or translating between sensors, no soil-moisture labels needed
- Output: *embeddings* — compact vectors that summarise what a patch of land looks like

Why for soil moisture

- In-situ labels are scarce ($\sim$1000 stations); pre-training brings knowledge from billions of pixels
- One shared representation for S2, S1, DEM and land cover

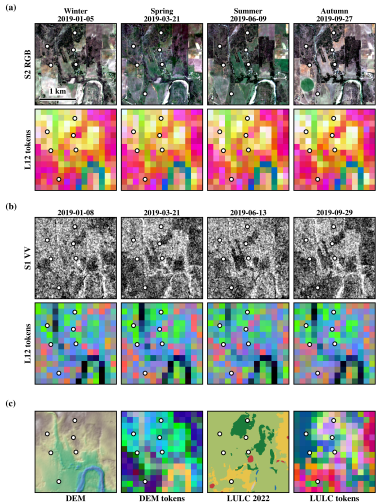
TerraMind (used here)

- Multi-modal, any-to-any GFM (IBM / ESA)
- ViT-Base: 12 layers, 768-dim, used *frozen*
- Input tile 2.24 km $\rightarrow$ $14 \times 14 = 196$ tokens, each a 160 m patch
- Modalities used: S2 L2A, S1 RTC, DEM, LULC
- Layers L3, L6, L9, L12 extracted

Open question

- Do these general-purpose embeddings carry soil-moisture information — or only describe the landscape?

Motivation TerraMind L12 tokens: what the model receives

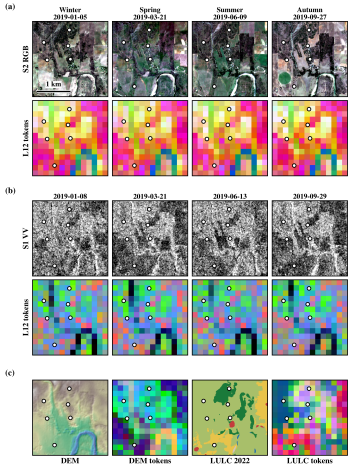


- (a) S2 RGB and its L12 tokens, four seasons; (b) S1 VV and tokens; (c) DEM, LULC and their tokens. Colours = first 3 principal components of the 768-dim tokens.
- Tokens follow fields, woodland and the river — the landscape structure is preserved at 160 m
- S2 tokens look almost the same in all four seasons: the embedding is dominated by *what the land is*, less by *how wet it is today*
- This is what RQ1 tests quantitatively

Eight research questions

Block	RQ	Question
Core	RQ1	Do frozen GFM embeddings add skill beyond a multimodal driver baseline?
	RQ2	What does the model generalise to — places or processes?
	RQ3	How does skill vary across the landscape?
Mechanism	RQ4	Which modalities matter, at which depth?
	RQ5	Does the fine-resolution path carry real sub-km information?
Negative	RQ6	Is land-surface temperature a useful auxiliary signal for SM?
Outlook	RQ7	Can the model detect management signals (irrigation)?
	RQ8	Does the representation transfer to an unseen region?

RQ1 Do frozen GFM embeddings add skill beyond a multimodal driver baseline?



TerraMind L12 tokens for S2, S1, DEM and LULC at one tile (TxSON, 2019).

Have

- Modality-shuffle ablation, §59 model, 221 OOS stations (next slide)
- Satellite tokens swapped across stations: +37 % ubRMSE, r 0.77 $\rightarrow$ 0.52

Missing

- Drivers-only model (10 epochs)
- Raw-imagery control (GFM vs pixels)
- Within-station S2 shuffle

RQ1 What does the model use the GFM embeddings for? (shuffle ablation)

	0–10 cm		10–30 cm		30–100 cm	
Shuffled input	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse
All satellite, cross	+36.8% (0.52)	97%	+37.1% (0.52)	96%	+38.4% (0.44)	92%
S2, cross	+4.9% (0.74)	84%	+6.8% (0.72)	87%	+8.7% (0.63)	85%
S1, cross	+4.8% (0.73)	82%	+7.7% (0.70)	84%	+14.4% (0.58)	86%
S1, within	0.0% (0.75)	49%	+1.0% (0.73)	59%	0.0% (0.64)	49%
DEM, cross	+12.6% (0.71)	90%	+9.7% (0.68)	86%	+18.4% (0.59)	96%
LULC, cross	+6.1% (0.73)	82%	+6.6% (0.70)	82%	+12.7% (0.62)	87%
Fine 20 m path, cross	+0.5% (0.75)	56%	+1.3% (0.74)	65%	+0.1% (0.68)	52%
ERA5, cross	+30.3% (0.50)	95%	+24.6% (0.54)	91%	+15.1% (0.50)	86%
ERA5, within	+39.0% (0.35)	88%	+30.4% (0.37)	81%	+13.5% (0.34)	73%
Soil, cross	+7.6% (0.72)	87%	+17.6% (0.63)	92%	+25.7% (0.55)	94%
SIF, cross	+1.3% (0.75)	63%	+1.6% (0.73)	62%	+1.8% (0.67)	71%
TWSA, cross	0.0% (0.77)	50%	0.0% (0.76)	53%	+0.1% (0.71)	58%

Baseline: ubRMSE 0.050 / 0.048 / 0.048, r 0.766 / 0.761 / 0.702 (n = 221 / 144 / 119). *Cross* = another station's input (kills site identity); *within* = same station, shuffled dates (kills timing).

RQ1 What we can and cannot claim today

Can claim

1. Reliance on GFM tokens $\approx$ reliance on weather (+37 % vs +30 %).
2. Tokens are a *static landscape descriptor*: S1 within-station shuffle = 0 %.
3. They shape the response, not just the mean (ubRMSE is bias-free): *GFM embeddings condition the weather$\rightarrow$SM transfer on the landscape.*
4. Depth pattern inverted vs a moisture signal: S1 +4.8 % $\rightarrow$ +14.4 % with depth.
5. Modalities redundant: S2+S1+DEM+LULC $\approx$ 28 % $<$ 37 % joint.
6. Fine 20 m path and TWSA unused.

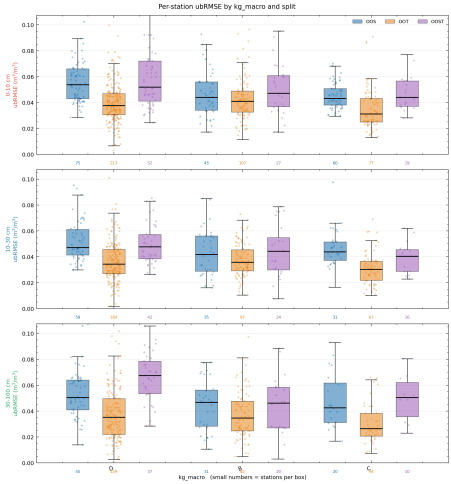
Cannot claim yet

- *Added* skill — a shuffle measures reliance
- GFM better than raw imagery
- Anything about S2 dynamics

To answer fully

- Within-station S2 shuffle (eval only)
- Run A: drivers + soil, no tokens
- Run B: A + static site descriptors
- Full $>$ B $>$ A $\Rightarrow$ dynamic skill;
  Full $\approx$ B $>$ A $\Rightarrow$ site info only

RQ2 What does the model generalise to — places or processes?



Per-station ubRMSE by climate zone and split (OOS / OOT / OOST).

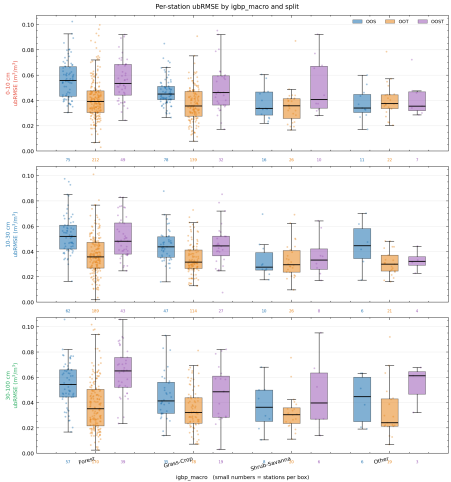
Have

- Splits §47: train 573 / val 89 / OOS 224; OOT 2023–2025
- OOT error is *lower* than OOS in every climate zone: new years at known sites are easy, new sites are hard
- `pw_stage2a_L3`: OOT 0.0434 beat val 0.0507
- Step-0 offsets probe: pair sign 0.50; site-fingerprint verdict confirmed (27572594)

Missing

- One val / OOT / OOS table for the selected model

RQ3 How does skill vary across the landscape?



Per-station ubRMSE by land-cover class and split.

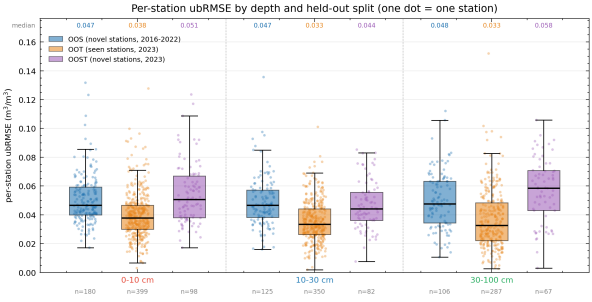
Have

- Per-station metrics by climate zone and land cover
- LULC, elevation band, soil, HAND for every station
- HAND stable under DEM noise; TWI is not — use HAND

Missing

- Error vs aridity, HAND, soil texture, per depth
- Flag elevation sentinels first (18 REMEDHUS = −99.9, Baytik = 0)

RQ4 Which modalities matter, at which depth?



Per-station ubRMSE by depth.

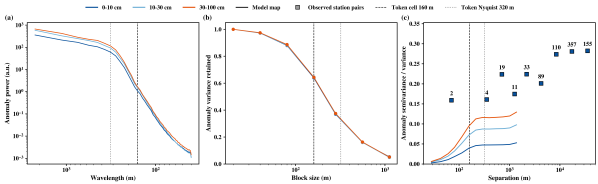
Have (shuffle ablation, RQ1 table)

- ERA5 dynamics dominate the surface (+39 %) and fade with depth (+14 %)
- Soil properties grow with depth (+7.6 % → +25.7 %)
- SIF ≈ 1.5 %; TWSA ≈ 0
- §58: 180-day ERA5 window

Hypothesis status

- "TWSA drives the root zone" — not supported so far

RQ5 Does the fine-resolution path carry real sub-kilometre information?



Effective-resolution test at TxSON (§70).

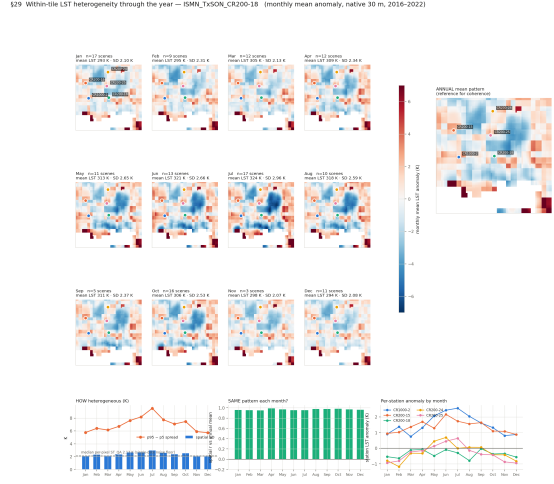
Have

- §53 fine ablation: SM ignores the 20 m path (LST reads it)
- Fine-path shuffle: +0.1 to +1.3 % ubRMSE
- §48 map flattens after epoch 8
- Overijssel grid run (§68, paused)

Framing

- A publishable negative result if framed well

RQ6 Is land-surface temperature a useful auxiliary signal for soil moisture?



Within-tile Landsat LST anomaly by month, TxSON CR200-18 (§29).

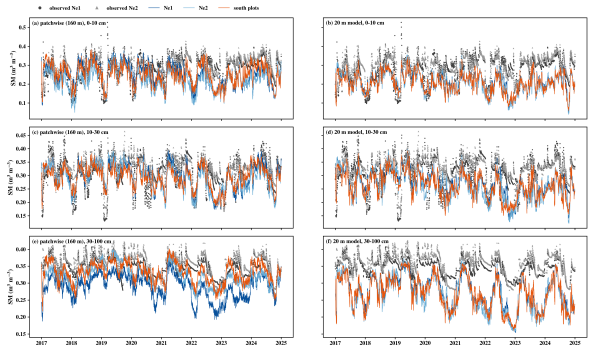
Have (all negative)

- Within-tile LST pattern is static: $r = +0.967$ across months
- Simpson's paradox: +0.167 pooled, −0.077 within station
- ECOSTRESS DTR = day pattern (r = +0.816); G0 fails
- In training LST hurts SM: val 0.0537 with vs 0.0487 without

Status

- Ready to write as a short paper / section

RQ7 Can the model detect management signals such as irrigation?



Mead Ne1/Ne2/Ne3: model vs observed SM, 2017–2024 (§69).

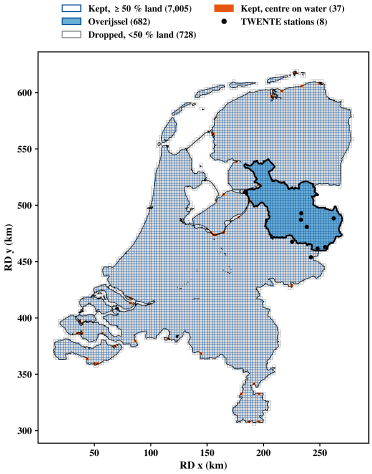
Have

- Mead daily SM built for Ne1/Ne2/Ne3
- Irrigation visible only in the dry year (2022)

Missing

- Irrigation log covers 2014–2021 only
- Prediction-minus-observation analysis vs irrigation dates

RQ8 Does the representation transfer to an unseen region?



Overijssel grid tiles (§68).

Have

- Overijssel: cloud mask done 688/688
- TWENTE: 8 stations, all OOS (end 2020, no OOT)

Missing

- Prep → tokenise → post → predict
- Map + TWENTE station comparison